

2.1

[illegible]

```
Activities Terminal Feb 16 14:26 seed@VM: ~/Labsetup
01;35:*.emf=01;35:*.ogv=01;35:*.ogx=01;35:*.aac=00;36:*.au=00;36:*.flac=00;36:*.m4a=00;36:*.mid=00;36:*.midi=00;36:*.mka=00;36:*.mp3=00;36:*.mpc=00;36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=00;36:*.opus=00;36:*.spx=00;36:*.xspf=00;36:
XDG_CURRENT_DESKTOP=ubuntu:GNOME
VTE_VERSION=6003
GNOME_TERMINAL_SCREEN=/org/gnome/Terminal/screen/3675cd9f_7506_4ebe_984d_3ecb95fba4bd
INVOCATION_ID=35af98fd94e04a60b3c9c6e4b7dfbda7
MANAGERPID=1752
LESSCLOSE=/usr/bin/lesspipe %s %s
XDG_SESSION_CLASS=user
TERM=xterm-256color
LESSOPEN=| /usr/bin/lesspipe %s
USER=seed
GNOME_TERMINAL_SERVICE=:1.108
DISPLAY=:0
SHLVL=1
QT_IM_MODULE=ibus
DBUS_STARTER_ADDRESS=unix:path=/run/user/1000/bus,guid=53f94007eab263fc139eeb4569936342
XDG_RUNTIME_DIR=/run/user/1000
JOURNAL_STREAM=8:33413
XDG_DATA_DIRS=/usr/share/ubuntu:/usr/local/share:/usr/share:/var/lib/snapd/desktop
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin:.
GDMSESSION=ubuntu
DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/1000/bus,guid=53f94007eab263fc139eeb4569936342
_/usr/bin/printenv
[02/16/26]seed@VM:~/Labsetup$ printenv PWD
/home/seed/CSIT 460/Labsetup
[02/16/26]seed@VM:~/Labsetup$ env | grep PWD
PWD=/home/seed/CSIT 460/Labsetup
[02/16/26]seed@VM:~/Labsetup$ export MYNAME="Duy"
[02/16/26]seed@VM:~/Labsetup$ printenv MYNAME
Duy
[02/16/26]seed@VM:~/Labsetup$

Activities Terminal Feb 16 14:28 seed@VM: ~/Labsetup
XDG_CURRENT_DESKTOP=ubuntu:GNOME
VTE_VERSION=6003
GNOME_TERMINAL_SCREEN=/org/gnome/Terminal/screen/3675cd9f_7506_4ebe_984d_3ecb95fba4bd
INVOCATION_ID=35af98fd94e04a60b3c9c6e4b7dfbda7
MANAGERPID=1752
LESSCLOSE=/usr/bin/lesspipe %s %s
XDG_SESSION_CLASS=user
TERM=xterm-256color
LESSOPEN=| /usr/bin/lesspipe %s
USER=seed
GNOME_TERMINAL_SERVICE=:1.108
DISPLAY=:0
SHLVL=1
QT_IM_MODULE=ibus
DBUS_STARTER_ADDRESS=unix:path=/run/user/1000/bus,guid=53f94007eab263fc139eeb4569936342
XDG_RUNTIME_DIR=/run/user/1000
JOURNAL_STREAM=8:33413
XDG_DATA_DIRS=/usr/share/ubuntu:/usr/local/share:/usr/share:/var/lib/snapd/desktop
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin:.
GDMSESSION=ubuntu
DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/1000/bus,guid=53f94007eab263fc139eeb4569936342
_/usr/bin/printenv
[02/16/26]seed@VM:~/Labsetup$ printenv PWD
/home/seed/CSIT 460/Labsetup
[02/16/26]seed@VM:~/Labsetup$ env | grep PWD
PWD=/home/seed/CSIT 460/Labsetup
[02/16/26]seed@VM:~/Labsetup$ export MYNAME="Duy"
[02/16/26]seed@VM:~/Labsetup$ printenv MYNAME
Duy
[02/16/26]seed@VM:~/Labsetup$ unset MYNAME
[02/16/26]seed@VM:~/Labsetup$ printenv MYNAME
[02/16/26]seed@VM:~/Labsetup$
```

In Task 1, I practiced viewing and modifying environment variables in Bash. To show the current environmental variables, I used the `printenv` and `env` commands, and I also viewed the `PWD` variable to show the working directory. Then, I used the `export` command to create a new variable called `MYNAME`, and I confirmed it was set correctly. Lastly, I used the `unset` command to delete the variable. This task is evidence of the fact that environment variables are shell-level settings that can be dynamically added or removed and are inherited by the programs started from the shell.

2.2

```
Activities Terminal Feb 16 14:36 seed@VM: ~/.../Labsetup
QT_IM_MODULE=ibus
DBUS_STARTER_ADDRESS=unix:path=/run/user/1000/bus,guid=53f94007eab263fc139eeb4569936342
XDG_RUNTIME_DIR=/run/user/1000
JOURNAL_STREAM=8:33413
XDG_DATA_DIRS=/usr/share/ubuntu:/usr/local/share:/usr/share:/var/lib/snapd/desktop
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin:.
GDMSESSION=ubuntu
DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/1000/bus,guid=53f94007eab263fc139eeb4569936342
=/usr/bin/printenv
[02/16/26]seed@VM:~/.../Labsetup$ printenv PWD
/home/seed/CSIT 460/Labsetup
[02/16/26]seed@VM:~/.../Labsetup$ env | grep PWD
PWD=/home/seed/CSIT 460/Labsetup
[02/16/26]seed@VM:~/.../Labsetup$ export MYNAME="Duy"
[02/16/26]seed@VM:~/.../Labsetup$ printenv MYNAME
Duy
[02/16/26]seed@VM:~/.../Labsetup$ unset MYNAME
[02/16/26]seed@VM:~/.../Labsetup$ printenv MYNAME
[02/16/26]seed@VM:~/.../Labsetup$ gcc myprintenv.c -o myprintenv
[02/16/26]seed@VM:~/.../Labsetup$ ./myprintenv > child.txt
[02/16/26]seed@VM:~/.../Labsetup$ head child.txt
SHELL=/bin/bash
SESSION_MANAGER=local/VM:@/tmp/.ICE-unix/1992,unix/VM:/tmp/.ICE-unix/1992
QT_ACCESSIBILITY=1
COLORTERM=truecolor
XDG_CONFIG_DIRS=/etc/xdg/xdg-ubuntu:/etc/xdg
XDG_MENU_PREFIX=gnome-
GNOME_DESKTOP_SESSION_ID=this-is-deprecated
GNOME_SHELL_SESSION_MODE=ubuntu
SSH_AUTH_SOCK=/run/user/1000/keyring/ssh
XMODIFIERS=@im=ibus
[02/16/26]seed@VM:~/.../Labsetup$

Activities Terminal Feb 16 14:40 seed@VM: ~/.../Labsetup
[02/16/26]seed@VM:~/.../Labsetup$ printenv MYNAME
Duy
[02/16/26]seed@VM:~/.../Labsetup$ unset MYNAME
[02/16/26]seed@VM:~/.../Labsetup$ printenv MYNAME
[02/16/26]seed@VM:~/.../Labsetup$ gcc myprintenv.c -o myprintenv
[02/16/26]seed@VM:~/.../Labsetup$ ./myprintenv > child.txt
[02/16/26]seed@VM:~/.../Labsetup$ head child.txt
SHELL=/bin/bash
SESSION_MANAGER=local/VM:@/tmp/.ICE-unix/1992,unix/VM:/tmp/.ICE-unix/1992
QT_ACCESSIBILITY=1
COLORTERM=truecolor
XDG_CONFIG_DIRS=/etc/xdg/xdg-ubuntu:/etc/xdg
XDG_MENU_PREFIX=gnome-
GNOME_DESKTOP_SESSION_ID=this-is-deprecated
GNOME_SHELL_SESSION_MODE=ubuntu
SSH_AUTH_SOCK=/run/user/1000/keyring/ssh
XMODIFIERS=@im=ibus
[02/16/26]seed@VM:~/.../Labsetup$ nano myprintenv.c
[02/16/26]seed@VM:~/.../Labsetup$ gcc myprintenv.c -o myprintenv
[02/16/26]seed@VM:~/.../Labsetup$ ./myprintenv > parent.txt
[02/16/26]seed@VM:~/.../Labsetup$ head parent.txt
SHELL=/bin/bash
SESSION_MANAGER=local/VM:@/tmp/.ICE-unix/1992,unix/VM:/tmp/.ICE-unix/1992
QT_ACCESSIBILITY=1
COLORTERM=truecolor
XDG_CONFIG_DIRS=/etc/xdg/xdg-ubuntu:/etc/xdg
XDG_MENU_PREFIX=gnome-
GNOME_DESKTOP_SESSION_ID=this-is-deprecated
GNOME_SHELL_SESSION_MODE=ubuntu
SSH_AUTH_SOCK=/run/user/1000/keyring/ssh
XMODIFIERS=@im=ibus
[02/16/26]seed@VM:~/.../Labsetup$
```

```
Activities Terminal Feb 16 14:43 seed@VM: ~/.../Labsetup
Duy
[02/16/26]seed@VM:~/.../Labsetup$ unset MYNAME
[02/16/26]seed@VM:~/.../Labsetup$ printenv MYNAME
[02/16/26]seed@VM:~/.../Labsetup$ gcc myprintenv.c -o myprintenv
[02/16/26]seed@VM:~/.../Labsetup$ ./myprintenv > child.txt
[02/16/26]seed@VM:~/.../Labsetup$ head child.txt
SHELL=/bin/bash
SESSION_MANAGER=local/VM:@/tmp/.ICE-unix/1992,unix/VM:/tmp/.ICE-unix/1992
QT_ACCESSIBILITY=1
COLORTERM=truecolor
XDG_CONFIG_DIRS=/etc/xdg/xdg-ubuntu:/etc/xdg
XDG_MENU_PREFIX=gnome-
GNOME_DESKTOP_SESSION_ID=this-is-deprecated
GNOME_SHELL_SESSION_MODE=ubuntu
SSH_AUTH_SOCK=/run/user/1000/keyring/ssh
XMODIFIERS=@im=ibus
[02/16/26]seed@VM:~/.../Labsetup$ nano myprintenv.c
[02/16/26]seed@VM:~/.../Labsetup$ gcc myprintenv.c -o myprintenv
[02/16/26]seed@VM:~/.../Labsetup$ ./myprintenv > parent.txt
[02/16/26]seed@VM:~/.../Labsetup$ head parent.txt
SHELL=/bin/bash
SESSION_MANAGER=local/VM:@/tmp/.ICE-unix/1992,unix/VM:/tmp/.ICE-unix/1992
QT_ACCESSIBILITY=1
COLORTERM=truecolor
XDG_CONFIG_DIRS=/etc/xdg/xdg-ubuntu:/etc/xdg
XDG_MENU_PREFIX=gnome-
GNOME_DESKTOP_SESSION_ID=this-is-deprecated
GNOME_SHELL_SESSION_MODE=ubuntu
SSH_AUTH_SOCK=/run/user/1000/keyring/ssh
XMODIFIERS=@im=ibus
[02/16/26]seed@VM:~/.../Labsetup$ diff child.txt parent.txt
[02/16/26]seed@VM:~/.../Labsetup$
```

In Task 2, I examined the issue of a child process getting environment variables from its parent using `fork()`. In this first step, the child process was instructed to print the environment variables and then it saved them to the `child.txt` file. For step 2, I reversed the code so now the parent can print out and save its own environment variables into the `parent.txt` file. I then compared the two files with the command `diff` and found that the outputs were the same. This is evidence of the fact that when a process forks the child process gains the environment variables of the parent process.

2.3

```
Activities Terminal Feb 16 14:50 seed@VM: ~/.../Labsetup
[02/16/26]seed@VM:~/.../Labsetup$ printenv MYNAME
[02/16/26]seed@VM:~/.../Labsetup$ gcc myprintenv.c -o myprintenv
[02/16/26]seed@VM:~/.../Labsetup$ ./myprintenv > child.txt
[02/16/26]seed@VM:~/.../Labsetup$ head child.txt
SHELL=/bin/bash
SESSION_MANAGER=local/VM:@/tmp/.ICE-unix/1992,unix/VM:/tmp/.ICE-unix/1992
QT_ACCESSIBILITY=1
COLORTERM=truecolor
XDG_CONFIG_DIRS=/etc/xdg/xdg-ubuntu:/etc/xdg
XDG_MENU_PREFIX=gnome-
GNOME_DESKTOP_SESSION_ID=this-is-deprecated
GNOME_SHELL_SESSION_MODE=ubuntu
SSH_AUTH_SOCK=/run/user/1000/keyring/ssh
XMODIFIERS=@im=ibus
[02/16/26]seed@VM:~/.../Labsetup$ nano myprintenv.c
[02/16/26]seed@VM:~/.../Labsetup$ gcc myprintenv.c -o myprintenv
[02/16/26]seed@VM:~/.../Labsetup$ ./myprintenv > parent.txt
[02/16/26]seed@VM:~/.../Labsetup$ head parent.txt
SHELL=/bin/bash
SESSION_MANAGER=local/VM:@/tmp/.ICE-unix/1992,unix/VM:/tmp/.ICE-unix/1992
QT_ACCESSIBILITY=1
COLORTERM=truecolor
XDG_CONFIG_DIRS=/etc/xdg/xdg-ubuntu:/etc/xdg
XDG_MENU_PREFIX=gnome-
GNOME_DESKTOP_SESSION_ID=this-is-deprecated
GNOME_SHELL_SESSION_MODE=ubuntu
SSH_AUTH_SOCK=/run/user/1000/keyring/ssh
XMODIFIERS=@im=ibus
[02/16/26]seed@VM:~/.../Labsetup$ diff child.txt parent.txt
[02/16/26]seed@VM:~/.../Labsetup$ gcc myenv.c -o myenv
[02/16/26]seed@VM:~/.../Labsetup$ ./myenv
[02/16/26]seed@VM:~/.../Labsetup$
```

```
Activities Terminal Feb 16 14:54 seed@VM: ~/.../Labsetup
XMODIFIERS=@im=ibus
[02/16/26]seed@VM:~/.../Labsetup$ nano myprintenv.c
[02/16/26]seed@VM:~/.../Labsetup$ gcc myprintenv.c -o myprintenv
[02/16/26]seed@VM:~/.../Labsetup$ ./myprintenv > parent.txt
[02/16/26]seed@VM:~/.../Labsetup$ head parent.txt
SHELL=/bin/bash
SESSION_MANAGER=local/VM:@/tmp/.ICE-unix/1992,unix/VM:/tmp/.ICE-unix/1992
QT_ACCESSIBILITY=1
COLORTERM=truecolor
XDG_CONFIG_DIRS=/etc/xdg/xdg-ubuntu:/etc/xdg
XDG_MENU_PREFIX=gnome-
GNOME_DESKTOP_SESSION_ID=this-is-deprecated
GNOME_SHELL_SESSION_MODE=ubuntu
SSH_AUTH_SOCK=/run/user/1000/keyring/ssh
XMODIFIERS=@im=ibus
[02/16/26]seed@VM:~/.../Labsetup$ diff child.txt parent.txt
[02/16/26]seed@VM:~/.../Labsetup$ gcc myenv.c -o myenv
[02/16/26]seed@VM:~/.../Labsetup$ ./myenv
[02/16/26]seed@VM:~/.../Labsetup$ nano myenv.c
[02/16/26]seed@VM:~/.../Labsetup$ gcc myenv.c -o myenv
[02/16/26]seed@VM:~/.../Labsetup$ ./myenv | head
SHELL=/bin/bash
SESSION_MANAGER=local/VM:@/tmp/.ICE-unix/1992,unix/VM:/tmp/.ICE-unix/1992
QT_ACCESSIBILITY=1
COLORTERM=truecolor
XDG_CONFIG_DIRS=/etc/xdg/xdg-ubuntu:/etc/xdg
XDG_MENU_PREFIX=gnome-
GNOME_DESKTOP_SESSION_ID=this-is-deprecated
GNOME_SHELL_SESSION_MODE=ubuntu
SSH_AUTH_SOCK=/run/user/1000/keyring/ssh
XMODIFIERS=@im=ibus
[02/16/26]seed@VM:~/.../Labsetup$
```

In Task 3, I carried out an experiment concerning the behavior of environment variables with the `execve()` phase of the system calls. When I executed the program with `execve("/usr/bin/env", argv, NULL)`, it printed almost no environment variables. But when I called `execve("/usr/bin/env", argv, environ)`, the output showed all the environment variables. This observation shows that a program run via `execve()` gets environment variables only if they are given specifically in the third argument.

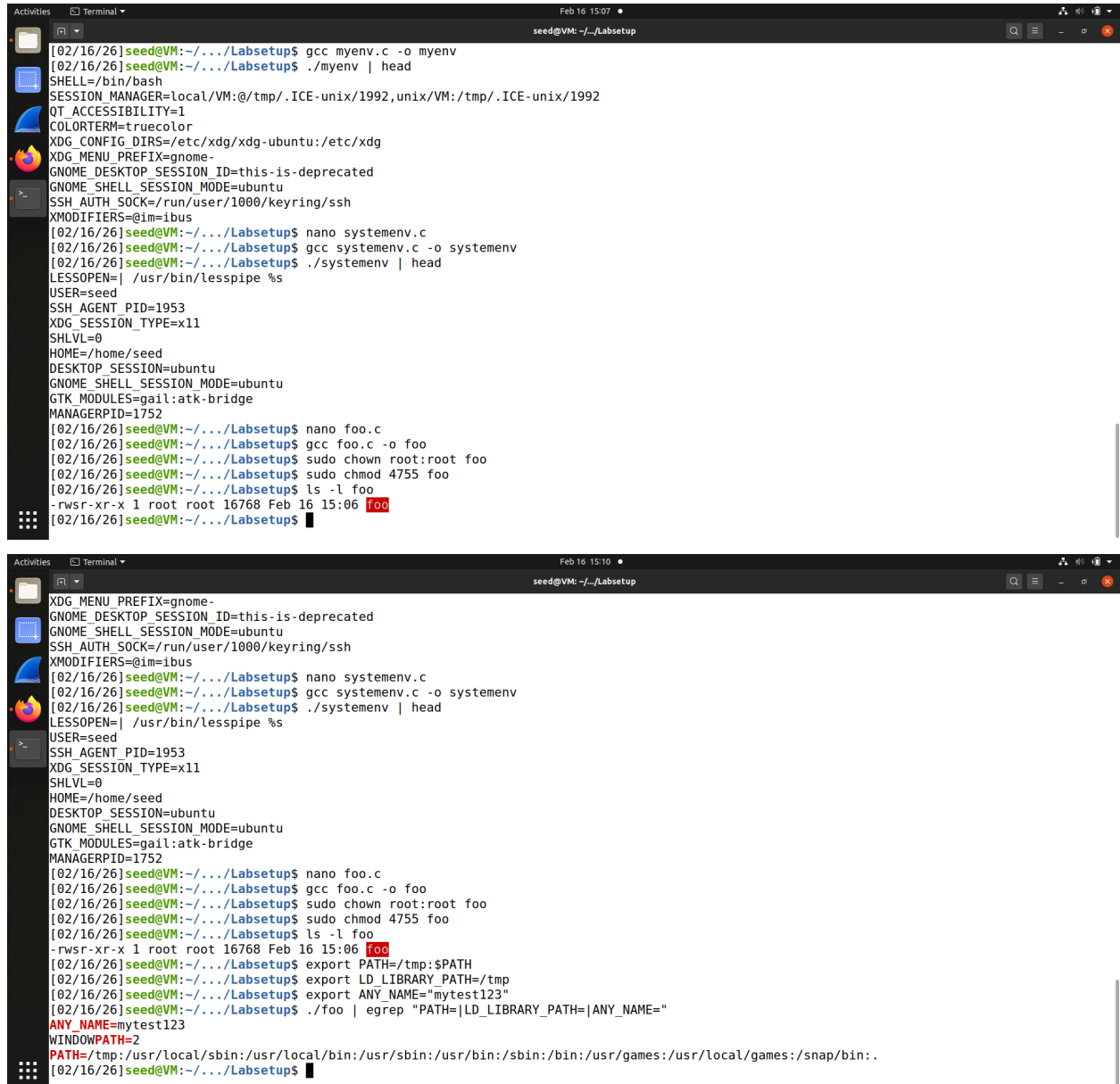
2.4

```
Activities Terminal Feb 16 15:01 seed@VM: ~/.../Labsetup
SSH_AUTH_SOCK=/run/user/1000/keyring/ssh
XMODIFIERS=@im=ibus
[02/16/26]seed@VM:~/.../Labsetup$ diff child.txt parent.txt
[02/16/26]seed@VM:~/.../Labsetup$ gcc myenv.c -o myenv
[02/16/26]seed@VM:~/.../Labsetup$ ./myenv
[02/16/26]seed@VM:~/.../Labsetup$ nano myenv.c
[02/16/26]seed@VM:~/.../Labsetup$ gcc myenv.c -o myenv
[02/16/26]seed@VM:~/.../Labsetup$ ./myenv | head
SHELL=/bin/bash
SESSION_MANAGER=local/VM:@/tmp/.ICE-unix/1992,unix/VM:/tmp/.ICE-unix/1992
QT_ACCESSIBILITY=1
COLORTERM=truecolor
XDG_CONFIG_DIRS=/etc/xdg/xdg-ubuntu:/etc/xdg
XDG_MENU_PREFIX=gnome-
GNOME_DESKTOP_SESSION_ID=this-is-deprecated
GNOME_SHELL_SESSION_MODE=ubuntu
SSH_AUTH_SOCK=/run/user/1000/keyring/ssh
XMODIFIERS=@im=ibus
[02/16/26]seed@VM:~/.../Labsetup$ nano systemenv.c
[02/16/26]seed@VM:~/.../Labsetup$ gcc systemenv.c -o systemenv
[02/16/26]seed@VM:~/.../Labsetup$ ./systemenv | head
LESSOPEN=| /usr/bin/lesspipe %s
USER=seed
SSH_AGENT_PID=1953
XDG_SESSION_TYPE=x11
SHLVL=0
HOME=/home/seed
DESKTOP_SESSION=ubuntu
GNOME_SHELL_SESSION_MODE=ubuntu
GTK_MODULES=gail:atk-bridge
MANAGERPID=1752
[02/16/26]seed@VM:~/.../Labsetup$
```

In Task 4, I did an experiment to see what happens to environment variables when a new program is executed with the use of the `system()` function. Unlike `execve()`, which launches the program directly, `system()` first starts a shell (`/bin/sh`) and then instructs the shell to run the

command. My compilation and execution of the program using `system("/usr/bin/env")` resulted in a display of the complete environment variable list. This shows the idea that `system()` leads the calling process to send its environment variables into the new shell.

2.5



```
[02/16/26]seed@VM:~/.../Labsetup$ gcc myenv.c -o myenv
[02/16/26]seed@VM:~/.../Labsetup$ ./myenv | head
SHELL=/bin/bash
SESSION_MANAGER=local/VM:@/tmp/.ICE-unix/1992,unix/VM:/tmp/.ICE-unix/1992
QT_ACCESSIBILITY=1
COLORTERM=truecolor
XDG_CONFIG_DIRS=/etc/xdg/xdg-ubuntu:/etc/xdg
XDG_MENU_PREFIX=gnome-
GNOME_DESKTOP_SESSION_ID=this-is-deprecated
GNOME_SHELL_SESSION_MODE=ubuntu
SSH_AUTH_SOCK=/run/user/1000/keyring/ssh
XMODIFIERS=@im=ibus
[02/16/26]seed@VM:~/.../Labsetup$ nano systemenv.c
[02/16/26]seed@VM:~/.../Labsetup$ gcc systemenv.c -o systemenv
[02/16/26]seed@VM:~/.../Labsetup$ ./systemenv | head
LESSOPEN=| /usr/bin/lesspipe %s
USER=seed
SSH_AGENT_PID=1953
XDG_SESSION_TYPE=x11
SHLVL=0
HOME=/home/seed
DESKTOP_SESSION=ubuntu
GNOME_SHELL_SESSION_MODE=ubuntu
GTK_MODULES=gail:atk-bridge
MANAGERPID=1752
[02/16/26]seed@VM:~/.../Labsetup$ nano foo.c
[02/16/26]seed@VM:~/.../Labsetup$ gcc foo.c -o foo
[02/16/26]seed@VM:~/.../Labsetup$ sudo chown root:root foo
[02/16/26]seed@VM:~/.../Labsetup$ sudo chmod 4755 foo
[02/16/26]seed@VM:~/.../Labsetup$ ls -l foo
-rwsr-xr-x 1 root root 16768 Feb 16 15:06 foo
[02/16/26]seed@VM:~/.../Labsetup$
```

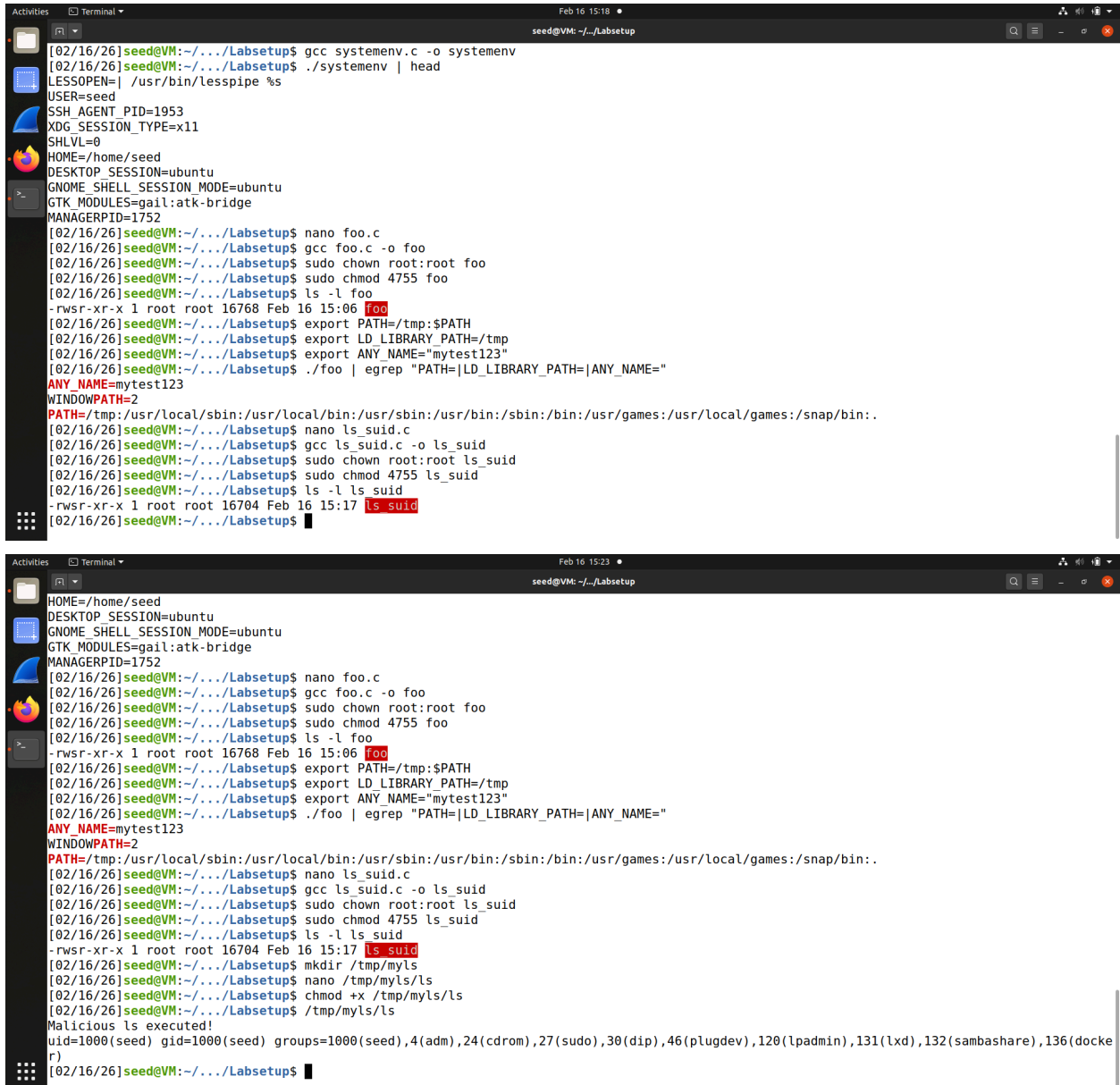


```
XDG_MENU_PREFIX=gnome-
GNOME_DESKTOP_SESSION_ID=this-is-deprecated
GNOME_SHELL_SESSION_MODE=ubuntu
SSH_AUTH_SOCK=/run/user/1000/keyring/ssh
XMODIFIERS=@im=ibus
[02/16/26]seed@VM:~/.../Labsetup$ nano systemenv.c
[02/16/26]seed@VM:~/.../Labsetup$ gcc systemenv.c -o systemenv
[02/16/26]seed@VM:~/.../Labsetup$ ./systemenv | head
LESSOPEN=| /usr/bin/lesspipe %s
USER=seed
SSH_AGENT_PID=1953
XDG_SESSION_TYPE=x11
SHLVL=0
HOME=/home/seed
DESKTOP_SESSION=ubuntu
GNOME_SHELL_SESSION_MODE=ubuntu
GTK_MODULES=gail:atk-bridge
MANAGERPID=1752
[02/16/26]seed@VM:~/.../Labsetup$ nano foo.c
[02/16/26]seed@VM:~/.../Labsetup$ gcc foo.c -o foo
[02/16/26]seed@VM:~/.../Labsetup$ sudo chown root:root foo
[02/16/26]seed@VM:~/.../Labsetup$ sudo chmod 4755 foo
[02/16/26]seed@VM:~/.../Labsetup$ ls -l foo
-rwsr-xr-x 1 root root 16768 Feb 16 15:06 foo
[02/16/26]seed@VM:~/.../Labsetup$ export PATH=/tmp:$PATH
[02/16/26]seed@VM:~/.../Labsetup$ export LD_LIBRARY_PATH=/tmp
[02/16/26]seed@VM:~/.../Labsetup$ export ANY_NAME="mytest123"
[02/16/26]seed@VM:~/.../Labsetup$ ./foo | egrep "PATH=|LD_LIBRARY_PATH=|ANY_NAME="
ANY_NAME=mytest123
WINDOWPATH=2
PATH=/tmp:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin:
[02/16/26]seed@VM:~/.../Labsetup$
```

In Task 5, I wrote a program (`foo`) that prints all environment variables through the use of the `environ` array. First, I compiled it and then, I turned it into a Set-UID root program via `chown root` and `chmod 4755`. After that, in my normal user shell, I set `PATH`, `LD_LIBRARY_PATH` and a custom variable `ANY_NAME`, and executed the setUID program. I found that my custom variable and `PATH` were displayed in the program's output. However, I was surprised that

LD_LIBRARY_PATH was not seen. This is because Linux sanitizes certain LD_* environment variables that are for SetUID programs to fight off library injection attacks.

2.6



```
[02/16/26]seed@VM:~/.../Labsetup$ gcc systemenv.c -o systemenv
[02/16/26]seed@VM:~/.../Labsetup$ ./systemenv | head
LESSOPEN=| /usr/bin/lesspipe %s
USER=seed
SSH_AGENT_PID=1953
XDG_SESSION_TYPE=x11
SHLVL=0
HOME=/home/seed
DESKTOP_SESSION=ubuntu
GNOME_SHELL_SESSION_MODE=ubuntu
GTK_MODULES=gail:atk-bridge
MANAGERPID=1752
[02/16/26]seed@VM:~/.../Labsetup$ nano foo.c
[02/16/26]seed@VM:~/.../Labsetup$ gcc foo.c -o foo
[02/16/26]seed@VM:~/.../Labsetup$ sudo chown root:root foo
[02/16/26]seed@VM:~/.../Labsetup$ sudo chmod 4755 foo
[02/16/26]seed@VM:~/.../Labsetup$ ls -l foo
-rwsr-xr-x 1 root root 16768 Feb 16 15:06 foo
[02/16/26]seed@VM:~/.../Labsetup$ export PATH=/tmp:$PATH
[02/16/26]seed@VM:~/.../Labsetup$ export LD_LIBRARY_PATH=/tmp
[02/16/26]seed@VM:~/.../Labsetup$ export ANY_NAME="mytest123"
[02/16/26]seed@VM:~/.../Labsetup$ ./foo | egrep "PATH=|LD_LIBRARY_PATH=|ANY_NAME="
ANY_NAME=mytest123
WINDOWPATH=2
PATH=/tmp:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin:.
[02/16/26]seed@VM:~/.../Labsetup$ nano ls_suid.c
[02/16/26]seed@VM:~/.../Labsetup$ gcc ls_suid.c -o ls_suid
[02/16/26]seed@VM:~/.../Labsetup$ sudo chown root:root ls_suid
[02/16/26]seed@VM:~/.../Labsetup$ sudo chmod 4755 ls_suid
[02/16/26]seed@VM:~/.../Labsetup$ ls -l ls_suid
-rwsr-xr-x 1 root root 16704 Feb 16 15:17 ls_suid
[02/16/26]seed@VM:~/.../Labsetup$
```

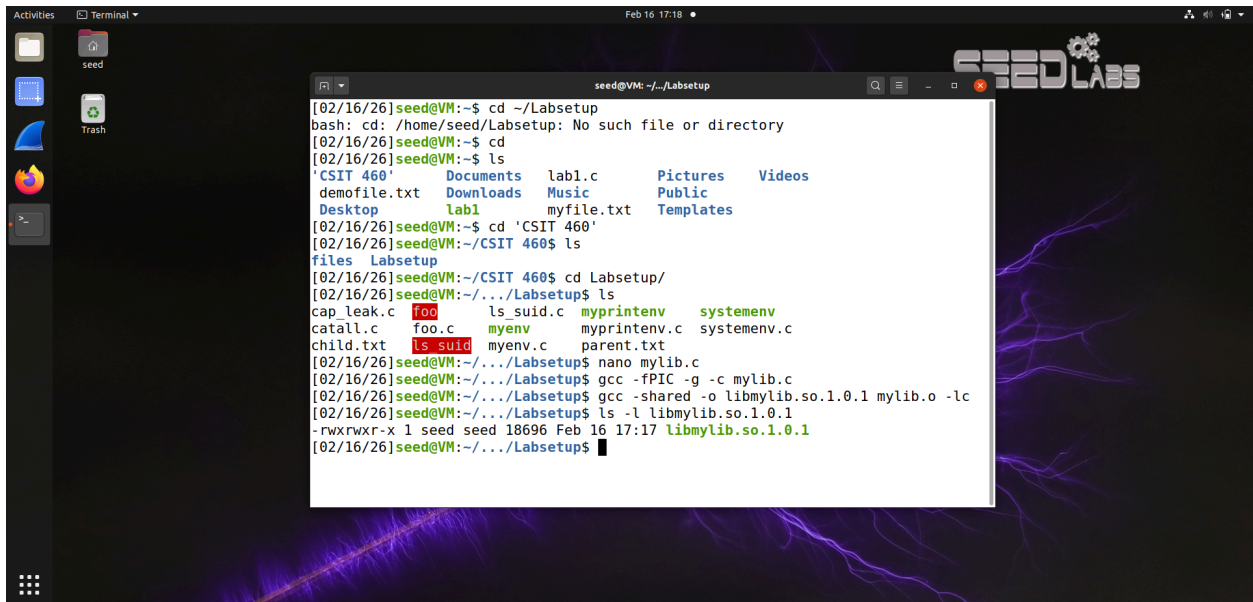
```
HOME=/home/seed
DESKTOP_SESSION=ubuntu
GNOME_SHELL_SESSION_MODE=ubuntu
GTK_MODULES=gail:atk-bridge
MANAGERPID=1752
[02/16/26]seed@VM:~/.../Labsetup$ nano foo.c
[02/16/26]seed@VM:~/.../Labsetup$ gcc foo.c -o foo
[02/16/26]seed@VM:~/.../Labsetup$ sudo chown root:root foo
[02/16/26]seed@VM:~/.../Labsetup$ sudo chmod 4755 foo
[02/16/26]seed@VM:~/.../Labsetup$ ls -l foo
-rwsr-xr-x 1 root root 16768 Feb 16 15:06 foo
[02/16/26]seed@VM:~/.../Labsetup$ export PATH=/tmp:$PATH
[02/16/26]seed@VM:~/.../Labsetup$ export LD_LIBRARY_PATH=/tmp
[02/16/26]seed@VM:~/.../Labsetup$ export ANY_NAME="mytest123"
[02/16/26]seed@VM:~/.../Labsetup$ ./foo | egrep "PATH=|LD_LIBRARY_PATH=|ANY_NAME="
ANY_NAME=mytest123
WINDOWPATH=2
PATH=/tmp:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin:.
[02/16/26]seed@VM:~/.../Labsetup$ nano ls_suid.c
[02/16/26]seed@VM:~/.../Labsetup$ gcc ls_suid.c -o ls_suid
[02/16/26]seed@VM:~/.../Labsetup$ sudo chown root:root ls_suid
[02/16/26]seed@VM:~/.../Labsetup$ sudo chmod 4755 ls_suid
[02/16/26]seed@VM:~/.../Labsetup$ ls -l ls_suid
-rwsr-xr-x 1 root root 16704 Feb 16 15:17 ls_suid
[02/16/26]seed@VM:~/.../Labsetup$ mkdir /tmp/myls
[02/16/26]seed@VM:~/.../Labsetup$ nano /tmp/myls/ls
[02/16/26]seed@VM:~/.../Labsetup$ chmod +x /tmp/myls/ls
[02/16/26]seed@VM:~/.../Labsetup$ /tmp/myls/ls
Malicious ls executed!
uid=1000(seed) gid=1000(seed) groups=1000(seed),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),120(lpadmin),131(lxd),132(sambashare),136(docke
r)
[02/16/26]seed@VM:~/.../Labsetup$
```

```
Activities Terminal Feb 16 15:25 seed@VM: ~/.../Labsetup
GTK_MODULES=gail:atk-bridge
MANAGERPID=1752
[02/16/26]seed@VM:~/.../Labsetup$ nano foo.c
[02/16/26]seed@VM:~/.../Labsetup$ gcc foo.c -o foo
[02/16/26]seed@VM:~/.../Labsetup$ sudo chown root:root foo
[02/16/26]seed@VM:~/.../Labsetup$ sudo chmod 4755 foo
[02/16/26]seed@VM:~/.../Labsetup$ ls -l foo
-rwsr-xr-x 1 root root 16768 Feb 16 15:06 foo
[02/16/26]seed@VM:~/.../Labsetup$ export PATH=/tmp:$PATH
[02/16/26]seed@VM:~/.../Labsetup$ export LD_LIBRARY_PATH=/tmp
[02/16/26]seed@VM:~/.../Labsetup$ export ANY_NAME="mytest123"
[02/16/26]seed@VM:~/.../Labsetup$ ./foo | egrep "PATH=|LD_LIBRARY_PATH=|ANY_NAME="
ANY_NAME=mytest123
WINDOWPATH=2
PATH=/tmp:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin:.
[02/16/26]seed@VM:~/.../Labsetup$ nano ls_suid.c
[02/16/26]seed@VM:~/.../Labsetup$ gcc ls_suid.c -o ls_suid
[02/16/26]seed@VM:~/.../Labsetup$ sudo chown root:root ls_suid
[02/16/26]seed@VM:~/.../Labsetup$ sudo chmod 4755 ls_suid
[02/16/26]seed@VM:~/.../Labsetup$ ls -l ls_suid
-rwsr-xr-x 1 root root 16704 Feb 16 15:17 ls_suid
[02/16/26]seed@VM:~/.../Labsetup$ mkdir /tmp/myls
[02/16/26]seed@VM:~/.../Labsetup$ nano /tmp/myls/ls
[02/16/26]seed@VM:~/.../Labsetup$ chmod +x /tmp/myls/ls
[02/16/26]seed@VM:~/.../Labsetup$ /tmp/myls/ls
Malicious ls executed!
uid=1000(seed) gid=1000(seed) groups=1000(seed),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),120(lpadmin),131(lxd),132(sambashare),136(docke
r)
[02/16/26]seed@VM:~/.../Labsetup$ export PATH=/tmp/myls:$PATH
[02/16/26]seed@VM:~/.../Labsetup$ which ls
/tmp/myls/ls
[02/16/26]seed@VM:~/.../Labsetup$

Activities Terminal Feb 16 15:27 seed@VM: ~/.../Labsetup
[02/16/26]seed@VM:~/.../Labsetup$ export LD_LIBRARY_PATH=/tmp
[02/16/26]seed@VM:~/.../Labsetup$ export ANY_NAME="mytest123"
[02/16/26]seed@VM:~/.../Labsetup$ ./foo | egrep "PATH=|LD_LIBRARY_PATH=|ANY_NAME="
ANY_NAME=mytest123
WINDOWPATH=2
PATH=/tmp:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin:.
[02/16/26]seed@VM:~/.../Labsetup$ nano ls_suid.c
[02/16/26]seed@VM:~/.../Labsetup$ gcc ls_suid.c -o ls_suid
[02/16/26]seed@VM:~/.../Labsetup$ sudo chown root:root ls_suid
[02/16/26]seed@VM:~/.../Labsetup$ sudo chmod 4755 ls_suid
[02/16/26]seed@VM:~/.../Labsetup$ ls -l ls_suid
-rwsr-xr-x 1 root root 16704 Feb 16 15:17 ls_suid
[02/16/26]seed@VM:~/.../Labsetup$ mkdir /tmp/myls
[02/16/26]seed@VM:~/.../Labsetup$ nano /tmp/myls/ls
[02/16/26]seed@VM:~/.../Labsetup$ chmod +x /tmp/myls/ls
[02/16/26]seed@VM:~/.../Labsetup$ /tmp/myls/ls
Malicious ls executed!
uid=1000(seed) gid=1000(seed) groups=1000(seed),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),120(lpadmin),131(lxd),132(sambashare),136(docke
r)
[02/16/26]seed@VM:~/.../Labsetup$ export PATH=/tmp/myls:$PATH
[02/16/26]seed@VM:~/.../Labsetup$ which ls
/tmp/myls/ls
[02/16/26]seed@VM:~/.../Labsetup$ ./ls_suid
Malicious ls executed!
uid=1000(seed) gid=1000(seed) groups=1000(seed),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),120(lpadmin),131(lxd),132(sambashare),136(docke
r)
[02/16/26]seed@VM:~/.../Labsetup$ sudo ln -sf /bin/zsh /bin/sh
[02/16/26]seed@VM:~/.../Labsetup$ ./ls_suid
Malicious ls executed!
uid=1000(seed) gid=1000(seed) groups=1000(seed),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),120(lpadmin),131(lxd),132(sambashare),136(docke
r)
[02/16/26]seed@VM:~/.../Labsetup$
```

During Task 6, I explored the impact of PATH environment variable on a SetUID-root program. The program has the command system("ls") which gives the indication that the program is not calling /bin/ls directly but is giving the command to a shell that first searches for ls in the path using the variable PATH. I wrote a malicious fake ls script in /tmp/myls and altered the PATH variable such that this folder would be searched before anything else. When I ran the SetUID program, it executed the ls script that I created in the first place instead of the real /bin/ls. The protection was bypassed after linking /bin/sh to zsh and the attack behavior got more evident. This is a clear example of why using system() in a privileged program can be dangerous.

2.7



```

[02/16/26]seed@VM:~$ cd ~/Labsetup
bash: cd: /home/seed/Labsetup: No such file or directory
[02/16/26]seed@VM:~$ cd
[02/16/26]seed@VM:~$ ls
'CSIT 460'  Documents  lab1.c      Pictures  Videos
demofile.txt Downloads  Music       Public
Desktop     lab1       myfile.txt  Templates
[02/16/26]seed@VM:~$ cd 'CSIT 460'
[02/16/26]seed@VM:~/CSIT 460$ ls
files  Labsetup
[02/16/26]seed@VM:~/CSIT 460$ cd Labsetup/
[02/16/26]seed@VM:~/.../Labsetup$ ls
cap_leak.c  foo      ls_suid.c  myprintenv  systemenv
catall.c    foo.c    myenv      myprintenv.c systemenv.c
child.txt   ls_suid  myenv.c    parent.txt
[02/16/26]seed@VM:~/.../Labsetup$ nano mylib.c
[02/16/26]seed@VM:~/.../Labsetup$ gcc -fPIC -g -c mylib.c
[02/16/26]seed@VM:~/.../Labsetup$ gcc -shared -o libmylib.so.1.0.1 mylib.o -lc
[02/16/26]seed@VM:~/.../Labsetup$ ls -l libmylib.so.1.0.1
-rwxrwxr-x 1 seed seed 18696 Feb 16 17:17 libmylib.so.1.0.1
[02/16/26]seed@VM:~/.../Labsetup$

```



```

[02/16/26]seed@VM:~$ cd ~/Labsetup
bash: cd: /home/seed/Labsetup: No such file or directory
[02/16/26]seed@VM:~$ cd
[02/16/26]seed@VM:~$ ls
'CSIT 460'  Documents  lab1.c      Pictures  Videos
demofile.txt Downloads  Music       Public
Desktop     lab1       myfile.txt  Templates
[02/16/26]seed@VM:~$ cd 'CSIT 460'
[02/16/26]seed@VM:~/CSIT 460$ ls
files  Labsetup
[02/16/26]seed@VM:~/CSIT 460$ cd Labsetup/
[02/16/26]seed@VM:~/.../Labsetup$ ls
cap_leak.c  foo      ls_suid.c  myprintenv  systemenv
catall.c    foo.c    myenv      myprintenv.c systemenv.c
child.txt   ls_suid  myenv.c    parent.txt
[02/16/26]seed@VM:~/.../Labsetup$ nano mylib.c
[02/16/26]seed@VM:~/.../Labsetup$ gcc -fPIC -g -c mylib.c
[02/16/26]seed@VM:~/.../Labsetup$ gcc -shared -o libmylib.so.1.0.1 mylib.o -lc
[02/16/26]seed@VM:~/.../Labsetup$ ls -l libmylib.so.1.0.1
-rwxrwxr-x 1 seed seed 18696 Feb 16 17:17 libmylib.so.1.0.1
[02/16/26]seed@VM:~/.../Labsetup$ export LD_PRELOAD=./libmylib.so.1.0.1
[02/16/26]seed@VM:~/.../Labsetup$ printenv LD_PRELOAD
./libmylib.so.1.0.1
[02/16/26]seed@VM:~/.../Labsetup$

```

```
Activities Terminal Feb 16 17:23 seed@VM: ~/Labsetup
[02/16/26]seed@VM:~$ cd ~/Labsetup
bash: cd: /home/seed/Labsetup: No such file or directory
[02/16/26]seed@VM:~$ cd
[02/16/26]seed@VM:~$ ls
'CSIT 460' Documents lab1.c Pictures Videos
demofile.txt Downloads Music Public
Desktop lab1 myfile.txt Templates
[02/16/26]seed@VM:~$ cd 'CSIT 460'
[02/16/26]seed@VM:~/CSIT 460$ ls
files Labsetup
[02/16/26]seed@VM:~/CSIT 460$ cd Labsetup/
[02/16/26]seed@VM:~/../Labsetup$ ls
cap_leak.c foo ls_suid.c myprintenv systemenv
catall.c foo.c myenv myprintenv.c systemenv.c
child.txt ls_suid myenv.c parent.txt
[02/16/26]seed@VM:~/../Labsetup$ nano mylib.c
[02/16/26]seed@VM:~/../Labsetup$ gcc -fPIC -g -c mylib.c
[02/16/26]seed@VM:~/../Labsetup$ gcc -shared -o libmylib.so.1.0.1 mylib.o -lc
[02/16/26]seed@VM:~/../Labsetup$ ls -l libmylib.so.1.0.1
-rwxrwxr-x 1 seed seed 18696 Feb 16 17:17 libmylib.so.1.0.1
[02/16/26]seed@VM:~/../Labsetup$ export LD_PRELOAD=./libmylib.so.1.0.1
[02/16/26]seed@VM:~/../Labsetup$ printenv LD_PRELOAD
./libmylib.so.1.0.1
[02/16/26]seed@VM:~/../Labsetup$ nano myprog.c
[02/16/26]seed@VM:~/../Labsetup$ gcc myprog.c -o myprog
[02/16/26]seed@VM:~/../Labsetup$ chmod 755 myprog
[02/16/26]seed@VM:~/../Labsetup$ ./myprog
I am not sleeping!
[02/16/26]seed@VM:~/../Labsetup$
```

```
Activities Terminal Feb 16 17:25 seed@VM: ~/Labsetup
[02/16/26]seed@VM:~$ cd
[02/16/26]seed@VM:~$ ls
'CSIT 460' Documents lab1.c Pictures Videos
demofile.txt Downloads Music Public
Desktop lab1 myfile.txt Templates
[02/16/26]seed@VM:~$ cd 'CSIT 460'
[02/16/26]seed@VM:~/CSIT 460$ ls
files Labsetup
[02/16/26]seed@VM:~/CSIT 460$ cd Labsetup/
[02/16/26]seed@VM:~/../Labsetup$ ls
cap_leak.c foo ls_suid.c myprintenv systemenv
catall.c foo.c myenv myprintenv.c systemenv.c
child.txt ls_suid myenv.c parent.txt
[02/16/26]seed@VM:~/../Labsetup$ nano mylib.c
[02/16/26]seed@VM:~/../Labsetup$ gcc -fPIC -g -c mylib.c
[02/16/26]seed@VM:~/../Labsetup$ gcc -shared -o libmylib.so.1.0.1 mylib.o -lc
[02/16/26]seed@VM:~/../Labsetup$ ls -l libmylib.so.1.0.1
-rwxrwxr-x 1 seed seed 18696 Feb 16 17:17 libmylib.so.1.0.1
[02/16/26]seed@VM:~/../Labsetup$ export LD_PRELOAD=./libmylib.so.1.0.1
[02/16/26]seed@VM:~/../Labsetup$ printenv LD_PRELOAD
./libmylib.so.1.0.1
[02/16/26]seed@VM:~/../Labsetup$ nano myprog.c
[02/16/26]seed@VM:~/../Labsetup$ gcc myprog.c -o myprog
[02/16/26]seed@VM:~/../Labsetup$ chmod 755 myprog
[02/16/26]seed@VM:~/../Labsetup$ ./myprog
I am not sleeping!
[02/16/26]seed@VM:~/../Labsetup$ sudo chown root:root myprog
[02/16/26]seed@VM:~/../Labsetup$ sudo chmod 4755 myprog
[02/16/26]seed@VM:~/../Labsetup$ ls -l myprog
-rwsr-xr-x 1 root root 16696 Feb 16 17:22 myprog
[02/16/26]seed@VM:~/../Labsetup$ ./myprog
[02/16/26]seed@VM:~/../Labsetup$
```

In Task 7, the LD_PRELOAD environment variable changes the behavior of normal programs relative to Set-UID programs. To begin, I created a shared library that overrides the sleep() function from the standard C library. In place of actually sleeping, the specific function will print the message "I am not sleeping!". Subsequently, after compiling the library and setting LD_PRELOAD, I executed myprog as a normal user. The output established that the dynamic loader had loaded my custom library initially and thus the override had been successfully accomplished.

Then, I made myprog a Set-UID root program by changing its ownership to root and enabling the Set-UID permission bit. Running the same program again as a regular user, this time, the override did not work, and the program behaved normally without printing the message. The

reason for this was that Linux sanitizes and removes dangerous LD_* environment variables and remove dangerous LD_* environment variables, such as LD_PRELOAD, when executing Set-UID programs. This is a security measure that prevents unprivileged users from injecting shared libraries into the executable file of privileged programs. This task demonstrated why environment variables are restricted in Set-UID programs for the protection of the system security.

2.8

```
Activities Terminal Feb 16 17:40 seed@VM: ~/Labsetup
child.txt ls suid myenv.c parent.txt
[02/16/26]seed@VM:~/Labsetup$ nano mylib.c
[02/16/26]seed@VM:~/Labsetup$ gcc -fPIC -g -c mylib.c
[02/16/26]seed@VM:~/Labsetup$ gcc -shared -o libmylib.so.1.0.1 mylib.o -lc
[02/16/26]seed@VM:~/Labsetup$ ls -l libmylib.so.1.0.1
-rwxrwxr-x 1 seed seed 18696 Feb 16 17:17 libmylib.so.1.0.1
[02/16/26]seed@VM:~/Labsetup$ export LD_PRELOAD=./libmylib.so.1.0.1
[02/16/26]seed@VM:~/Labsetup$ printenv LD_PRELOAD
./libmylib.so.1.0.1
[02/16/26]seed@VM:~/Labsetup$ nano myprog.c
[02/16/26]seed@VM:~/Labsetup$ gcc myprog.c -o myprog
[02/16/26]seed@VM:~/Labsetup$ chmod 755 myprog
[02/16/26]seed@VM:~/Labsetup$ ./myprog
I am not sleeping!
[02/16/26]seed@VM:~/Labsetup$ sudo chown root:root myprog
[02/16/26]seed@VM:~/Labsetup$ sudo chmod 4755 myprog
[02/16/26]seed@VM:~/Labsetup$ ls -l myprog
-rwsr-xr-x 1 root root 16696 Feb 16 17:22 myprog
[02/16/26]seed@VM:~/Labsetup$ ./myprog
[02/16/26]seed@VM:~/Labsetup$ sudo -i
root@VM:~# cd
root@VM:~# exit
logout
[02/16/26]seed@VM:~/Labsetup$ ls
cap_leak.c child.txt foo.c ls suid myenv mylib.c myprintenv myprog parent.txt systemenv.c
catall.c root libmylib.so.1.0.1 ls_suid.c myenv.c mylib.o myprintenv.c myprog.c systemenv
[02/16/26]seed@VM:~/Labsetup$ gcc catall.c -o catall
[02/16/26]seed@VM:~/Labsetup$ sudo chown root:root catall
[02/16/26]seed@VM:~/Labsetup$ sudo chmod 4755 catall
[02/16/26]seed@VM:~/Labsetup$ ls -l catall
-rwsr-xr-x 1 root root 16928 Feb 16 17:37 catall
[02/16/26]seed@VM:~/Labsetup$

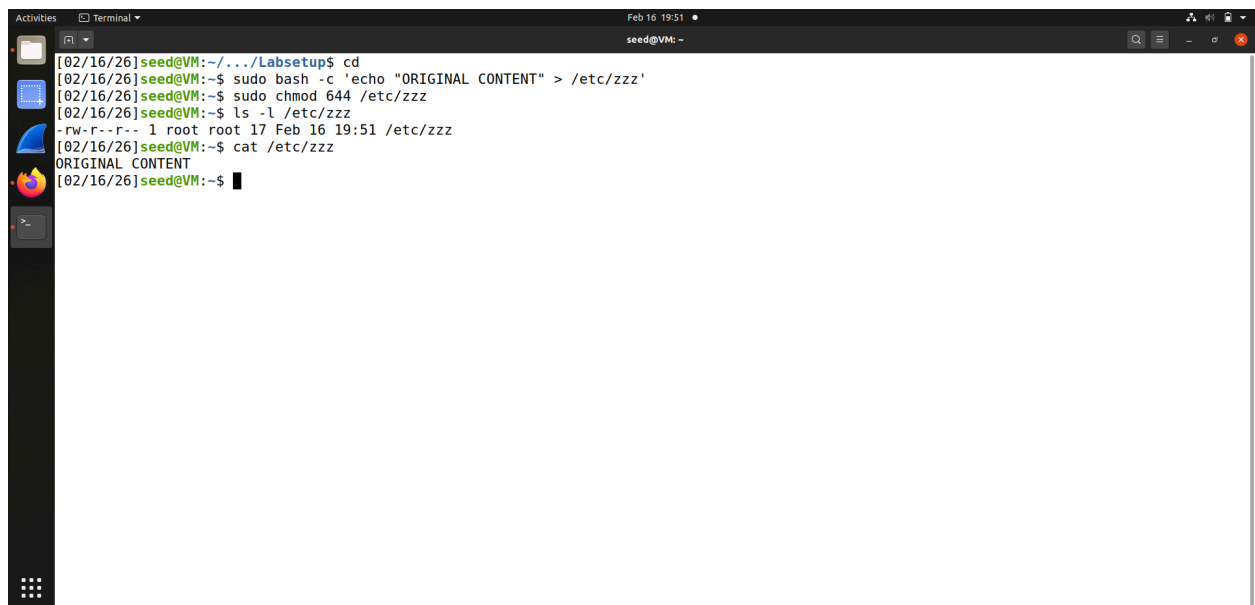
Activities Terminal Feb 16 18:06 seed@VM: ~/Labsetup
[02/16/26]seed@VM:~/Labsetup$ gcc myprog.c -o myprog
[02/16/26]seed@VM:~/Labsetup$ chmod 755 myprog
[02/16/26]seed@VM:~/Labsetup$ ./myprog
I am not sleeping!
[02/16/26]seed@VM:~/Labsetup$ sudo chown root:root myprog
[02/16/26]seed@VM:~/Labsetup$ sudo chmod 4755 myprog
[02/16/26]seed@VM:~/Labsetup$ ls -l myprog
-rwsr-xr-x 1 root root 16696 Feb 16 17:22 myprog
[02/16/26]seed@VM:~/Labsetup$ ./myprog
[02/16/26]seed@VM:~/Labsetup$ sudo -i
root@VM:~# cd
root@VM:~# exit
logout
[02/16/26]seed@VM:~/Labsetup$ ls
cap_leak.c child.txt foo.c ls suid myenv mylib.c myprintenv myprog parent.txt systemenv.c
catall.c root libmylib.so.1.0.1 ls_suid.c myenv.c mylib.o myprintenv.c myprog.c systemenv
[02/16/26]seed@VM:~/Labsetup$ gcc catall.c -o catall
[02/16/26]seed@VM:~/Labsetup$ sudo chown root:root catall
[02/16/26]seed@VM:~/Labsetup$ sudo chmod 4755 catall
[02/16/26]seed@VM:~/Labsetup$ ls -l catall
-rwsr-xr-x 1 root root 16928 Feb 16 17:37 catall
[02/16/26]seed@VM:~/Labsetup$ echo "hello" > test.txt
[02/16/26]seed@VM:~/Labsetup$ ./catall test.txt
hello
[02/16/26]seed@VM:~/Labsetup$ touch /tmp/importantfile
[02/16/26]seed@VM:~/Labsetup$ ls -l /tmp/importantfile
-rw-rw-r-- 1 seed seed 0 Feb 16 17:44 /tmp/importantfile
[02/16/26]seed@VM:~/Labsetup$ ./catall "test.txt; rm /tmp/importantfile"
hello
[02/16/26]seed@VM:~/Labsetup$ ls -l /tmp/importantfile
ls: cannot access '/tmp/importantfile': No such file or directory
[02/16/26]seed@VM:~/Labsetup$
```

```
Activities Terminal Feb 16 18:16 seed@VM: ~/Labsetup
root@VM:~# exit
logout
[02/16/26]seed@VM:~/../Labsetup$ ls
cap.leak.c child.txt foo.c ls.suid myenv mylib.c myprintenv myprog parent.txt systemenv.c
catall.c foo libmylib.so.1.0.1 ls.suid.c myenv.c mylib.o myprintenv.c myprog.c systemenv
[02/16/26]seed@VM:~/../Labsetup$ gcc catall.c -o catall
[02/16/26]seed@VM:~/../Labsetup$ sudo chown root:root catall
[02/16/26]seed@VM:~/../Labsetup$ sudo chmod 4755 catall
[02/16/26]seed@VM:~/../Labsetup$ ls -l catall
-rwsr-xr-x 1 root root 16928 Feb 16 17:37 catall
[02/16/26]seed@VM:~/../Labsetup$ echo "hello" > test.txt
[02/16/26]seed@VM:~/../Labsetup$ ./catall test.txt
hello
[02/16/26]seed@VM:~/../Labsetup$ touch /tmp/importantfile
[02/16/26]seed@VM:~/../Labsetup$ ls -l /tmp/importantfile
-rw-rw-r-- 1 seed seed 0 Feb 16 17:44 /tmp/importantfile
[02/16/26]seed@VM:~/../Labsetup$ ./catall "test.txt; rm /tmp/importantfile"
hello
[02/16/26]seed@VM:~/../Labsetup$ ls -l /tmp/importantfile
ls: cannot access '/tmp/importantfile': No such file or directory
[02/16/26]seed@VM:~/../Labsetup$ vim catall.c
[02/16/26]seed@VM:~/../Labsetup$ nano catall.c
[02/16/26]seed@VM:~/../Labsetup$ nano catall.c
[02/16/26]seed@VM:~/../Labsetup$ gcc catall.c -o catall
[02/16/26]seed@VM:~/../Labsetup$ sudo chown root:root catall
[02/16/26]seed@VM:~/../Labsetup$ sudo chmod 4755 catall
[02/16/26]seed@VM:~/../Labsetup$ touch /tmp/importantfile
[02/16/26]seed@VM:~/../Labsetup$ ./catall "test.txt; rm /tmp/importantfile"
/bin/cat: 'test.txt; rm /tmp/importantfile': No such file or directory
[02/16/26]seed@VM:~/../Labsetup$ ls -l /tmp/importantfile
-rw-rw-r-- 1 seed seed 0 Feb 16 18:13 /tmp/importantfile
[02/16/26]seed@VM:~/../Labsetup$
```

In Task 8, I examined the difference between `system()` and `execve()` for calling external programs in a SetUID-root program with elevated privileges. First, the program carried out the command `system` to display the contents of a file provided by the user. Due to the fact that `system()` runs a shell (`/bin/sh -c`), I could inject further commands like that with special shell characters in the input. For instance, I provided an argument to the effect `; rm ...`, which led to the running of both the `cat` command and a file deletion command thus compromising the integrity of the system.

Second, I replaced `system()` with `execve(v[0], v, NULL)`. After recompiling, the same attack no longer worked because `execve()` runs the program directly without invoking a shell. The input was read only as a filename rather than a command. This task consolidates the idea that using `system()` in SetUID programs is not secure because of the command injection risks but `execve()` offers a more secure approach to calling external programs.

2.9



```
Activities Terminal Feb 16 19:51 seed@VM: ~
[02/16/26]seed@VM:~/../Labsetup$ cd
[02/16/26]seed@VM:~$ sudo bash -c 'echo "ORIGINAL CONTENT" > /etc/zzz'
[02/16/26]seed@VM:~$ sudo chmod 644 /etc/zzz
[02/16/26]seed@VM:~$ ls -l /etc/zzz
-rw-r--r-- 1 root root 17 Feb 16 19:51 /etc/zzz
[02/16/26]seed@VM:~$ cat /etc/zzz
ORIGINAL CONTENT
[02/16/26]seed@VM:~$
```

```
Activities Terminal Feb 16 19:54 seed@VM: ~/.../Labsetup
[02/16/26]seed@VM:~$ cd
[02/16/26]seed@VM:~$ ls
[02/16/26]seed@VM:~$ cd 'CSIT 460'
[02/16/26]seed@VM:~/CSIT 460$ ls
files Labsetup
[02/16/26]seed@VM:~/CSIT 460$ cd Labsetup
[02/16/26]seed@VM:~/.../Labsetup$ gcc cap_leak.c -o cap_leak
[02/16/26]seed@VM:~/.../Labsetup$ sudo chown root:root cap_leak
[02/16/26]seed@VM:~/.../Labsetup$ sudo chmod 4755 cap_leak
[02/16/26]seed@VM:~/.../Labsetup$ ls -l cap_leak
-rwsr-xr-x 1 root root 17008 Feb 16 19:53 cap_leak
[02/16/26]seed@VM:~/.../Labsetup$
```

```
Activities Terminal Feb 16 20:02 seed@VM: ~/.../Labsetup
[02/16/26]seed@VM:~/.../Labsetup$ ./cap_leak
fd is 3
$ echo "Hacked by Duy" >&3
$ exit
[02/16/26]seed@VM:~/.../Labsetup$ cat /etc/zzz
ORIGINAL CONTENT
Hacked by Duy
[02/16/26]seed@VM:~/.../Labsetup$
```

In Task 9, I explored a vulnerability known as capability leaking in a SetUID-root program. The program opens the file `/etc/zzz` while it still has root privileges, and it receives a file descriptor pointing to this privileged file. Even though the program later drops its privileges using `setuid(getuid())`, it does not close the file descriptor. The program then executes a shell, and the shell inherits this open descriptor. As a result, I was able to write to `/etc/zzz` as a normal user by redirecting output to `/proc/$$/fd/X`, where `X` was the leaked descriptor number. This demonstrates that dropping root privileges is not enough if privileged resources such as open file descriptors are not properly cleaned up.

```
Activities Terminal Feb 16 19:55 seed@VM: ~/.../Labsetup
[02/16/26]seed@VM:~$ cd
[02/16/26]seed@VM:~$ ls
'CSIT 460' demofile.txt Desktop Documents Downloads lab1 lab1.c Music myfile.txt Pictures Public Templates Videos
[02/16/26]seed@VM:~$ cd 'CSIT 460'
[02/16/26]seed@VM:~/CSIT 460$ ls
files Labsetup
[02/16/26]seed@VM:~/CSIT 460$ cd Labsetup
[02/16/26]seed@VM:~/.../Labsetup$ gcc cap_leak.c -o cap_leak
[02/16/26]seed@VM:~/.../Labsetup$ sudo chown root:root cap_leak
[02/16/26]seed@VM:~/.../Labsetup$ sudo chmod 4755 cap_leak
[02/16/26]seed@VM:~/.../Labsetup$ ls -l cap_leak
-rwsr-xr-x 1 root root 17008 Feb 16 19:53 cap_leak
[02/16/26]seed@VM:~/.../Labsetup$ ./cap_leak
fd is 3
$
```